Astrophysical Fluid Dynamics

Lecture notes for ASTRO 442

David Radice

Contents

Preface . . . 7
How to use these notes . . . 7
Accessibility . . . 7
License . . . 7

Schedule . . . 9

I Foundations

1 Fluids as Continua . . . 13

2 Thermodynamics Refresher . . . 15

3 Core-Collapse Supernovae . . . 17

4 The Euler Equation . . . 19

5 Eulerian and Lagrangian Descriptions . . . 21

6 Conservation Laws . . . 23

7 The Gravitational Potential . . . 25

8 Hydrostatic Equilibrium and the Virial Theorem . . . 27
8.1 The virial theorem . . . 27
8.2 Hydrostatic equilibria . . . 29
8.3 Application to star formation . . . 30
8.4 In-class worksheet . . . 31
8.5 Exercises . . . 31
8.6 Further reading . . . 32

II Waves, Shocks, and Outflows

9 Sound Waves and the Doppler Effect . . . 35
9.1 Oscillation Equation . . . 35
9.2 Wave Energy and Momentum . . . 38
9.3 In-class worksheets . . . 41
9.3.1 Sound waves . . . 41
9.3.2 The Doppler effect . . . 41
9.4 Further reading . . . 41

10 Waves in Stratified Atmospheres . . . 43
10.1 Waves in Isothermal Atmospheres . . . 43
10.2 In-class worksheet . . . 46
10.3 Exercises . . . 46
10.4 Further reading . . . 47
11 Shock Waves . . . 49
12 Radiative Shocks . . . 51
13 Blast Waves . . . 53
14 Stellar Winds and Spherical Accretion . . . 55
III Instabilities, Viscosity, Disks, and Turbulence
15 Fluid Instabilities I: Convection . . . 59
16 Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz . . . 61
17 Fluid Instabilities III: Thermal and Gravitational Instabilities . . . 63
18 Diffusion and Viscous Flows . . . 65
19 Accretion Disks I: Structure and Viscous Evolution . . . 67
20 Accretion Disks II: Steady Disks and Spectra . . . 69
21 Turbulence . . . 71
IV Plasma and Magnetohydrodynamics
22 Introduction to Plasma . . . 75
23 The MHD Equations . . . 77
24 Flux Freezing . . . 79
25 Magnetic Tension and Hydromagnetic Equilibria . . . 81
26 MHD Waves . . . 83
27 The Magnetorotational Instability . . . 85
27.1 Qualitative Picture . . . 85
27.2 Detailed Calculation . . . 86
27.3 In-class worksheet . . . 89
27.4 Further reading . . . 89

Appendices **91**

A Vector Calculus **93**

Bibliography **95**

Preface

These are the lecture notes for ASTRO 442, an introduction to the fluid dynamics and magnetohydrodynamics of astrophysical systems. They are written to be self-contained: the textbook, Clarke and Carswell's *Principles of Astrophysical Fluid Dynamics* [1], is optional further reading. Each chapter ends with pointers to the relevant sections of the textbook.

> ! Work in progress
>
> I am in the process of importing my lecture notes from previous years. Once all chapters have been imported, the material will be revised and corrected in a second step, with added examples and clarifications. Until then, you may find typos, errors, and rough edges: please let me know when you spot one.
>
> Chapters marked "in preparation" will be completed before the corresponding lecture; see the Schedule.

How to use these notes

- **One chapter per lecture topic.** The Schedule lists which chapter goes with each lecture.
- **Everything is numbered by chapter.** Equation (9.14) is the fourteenth numbered equation of Chapter 9, and Exercise 8.4 is the fourth exercise of Chapter 8. When you ask a question about the notes, quote these numbers.
- **In-class worksheets** appear near the end of each chapter, followed by the **exercises**. Homework assignments on Canvas list the exercises to solve, for example "Exercises 2.1, 7.3".
- **Optional material** is set apart in boxes titled "Going further"; you can skip it on a first reading.

Accessibility

These notes are designed to meet the Web Content Accessibility Guidelines (WCAG) 2.1, level AA. Mathematics is rendered with MathJax, which supports screen readers, speech, and braille, and lets you explore an equation term by term: click on an equation or focus it with the keyboard, then use the arrow keys. Every figure has a text description.

If something in these notes is not accessible to you, please let me know.

License

The text and figures are licensed under CC BY-NC-SA 4.0. Code is licensed under the MIT license. The source is on GitHub.

Schedule

Lecture schedule for Spring 2027 and the chapter that goes with each lecture. Lectures are on Tuesdays and Thursdays. Homework assignments and due dates are posted on Canvas.

Lecture	Date	Topic	Notes
1	Tue Jan 12	Fluids as continua, continuity equation	Chapter 1.
2	Thu Jan 14	Energy equation, thermodynamics	Chapter 2.
3	Tue Jan 19	Core-collapse supernovae	Chapter 3.
4	Thu Jan 21	Vector calculus	Appendix A.
5	Tue Jan 26	Euler equation; Eulerian and Lagrangian descriptions	Chapter 4., Chapter 5.
6	Thu Jan 28	Conservation laws	Chapter 6.
7	Tue Feb 2	Gravitational potential	Chapter 7.
8	Thu Feb 4	Hydrostatic equilibrium, virial theorem, star formation	Chapter 8.
9	Tue Feb 9	Sound waves, acoustic power	Chapter 9.
10	Thu Feb 11	Doppler effect	Chapter 9.
11	Tue Feb 16	Sound waves in stratified media	Chapter 10.
12	Thu Feb 18	Shocks and shock adiabatics	Chapter 11.
13	Tue Feb 23	Radiative shocks	Chapter 12.
14	Thu Feb 25	Blast waves	Chapter 13.
15	Tue Mar 2	**Midterm I**	
16	Thu Mar 4	Stellar winds and spherical accretion	Chapter 14.
	Mar 9–11	*Spring break*	
17	Tue Mar 16	Fluid instabilities I	Chapter 15.
18	Thu Mar 18	Fluid instabilities II	Chapter 16.
19	Tue Mar 23	Introduction to plasma	Chapter 22.
20	Thu Mar 25	**Midterm II**	
21	Tue Mar 30	Fluid instabilities III	Chapter 17.

Lecture	Date	Topic	Notes
22	Thu Apr 1	Diffusion and viscous flows	Chapter 18.
23	Tue Apr 6	Accretion disks I	Chapter 19.
24	Thu Apr 8	Accretion disks II	Chapter 20.
25	Tue Apr 13	Turbulence	Chapter 21.
26	Thu Apr 15	MHD equations	Chapter 23.
27	Tue Apr 20	Flux freezing	Chapter 24.
28	Thu Apr 22	Magnetic tension	Chapter 25.
29	Tue Apr 27	MHD waves	Chapter 26.
30	Thu Apr 29	Magnetorotational instability	Chapter 27.

I Foundations

1 Fluids as Continua 13

2 Thermodynamics Refresher 15

3 Core-Collapse Supernovae 17

4 The Euler Equation 19

5 Eulerian and Lagrangian Descriptions 21

6 Conservation Laws 23

7 The Gravitational Potential 25

8 Hydrostatic Equilibrium and the Virial Theorem 27

8.1 The virial theorem 27
8.2 Hydrostatic equilibria 29
8.3 Application to star formation 30
8.4 In-class worksheet 31
8.5 Exercises 31
8.6 Further reading 32

1. Fluids as Continua

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

2. Thermodynamics Refresher

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

3. Core-Collapse Supernovae

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

4. The Euler Equation

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

5. Eulerian and Lagrangian Descriptions

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

6. Conservation Laws

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

7. The Gravitational Potential

> **Note**
>
> This chapter is in preparation.

8. Hydrostatic Equilibrium and the Virial Theorem

8.1 The virial theorem

We consider a self-gravitating cloud of gas in equilibrium, described by the momentum equation

$$\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}] = -\rho\nabla\Psi, \tag{8.1}$$

where Ψ is the gravitational potential. We consider the scalar moment of inertia[1]

$$I = \frac{1}{2}\int_V \rho(\mathbf{r})r^2\mathrm{d}V, \tag{8.2}$$

where V is a volume that encloses the whole gas cloud. The time derivative of I is (using the continuity equation)

$$\begin{aligned}\frac{\mathrm{d}I}{\mathrm{d}t} &= \frac{1}{2}\int_V \frac{\partial\rho}{\partial t}r^2\mathrm{d}V \\ &= -\frac{1}{2}\int_V \nabla\cdot(\rho\mathbf{v})r^2\mathrm{d}V \\ &= -\frac{1}{2}\int_V \nabla\cdot(\rho\mathbf{v}r^2)\mathrm{d}V + \frac{1}{2}\int_V \rho\mathbf{v}\cdot\nabla r^2\mathrm{d}V.\end{aligned} \tag{8.3}$$

The total divergence term is equal to a boundary integral

$$\int_V \nabla\cdot(\rho\mathbf{v}r^2)\mathrm{d}V = \int_S \rho\mathbf{v}r^2\cdot\mathbf{n}\mathrm{d}S, \tag{8.4}$$

but the latter vanishes since, by assumption, $\rho\to 0$ at the boundary of V. So we find

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{1}{2}\int_V \rho\mathbf{v}\cdot\nabla r^2\mathrm{d}V = \int_V \rho\mathbf{v}\cdot\mathbf{r}\mathrm{d}V. \tag{8.5}$$

Here, we have used the fact that $\nabla r = \mathbf{r}/r$.

The second time derivative of I is

$$\begin{aligned}\frac{\mathrm{d}^2I}{\mathrm{d}t^2} &= \int_V \frac{\partial(\rho\mathbf{v})}{\partial t}\cdot\mathbf{r}\mathrm{d}V \\ &= -\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}V \\ &\quad -\int_V \nabla\cdot[\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}]\cdot\mathbf{r}\mathrm{d}V \\ &= -\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}V - \int_V \nabla\cdot[\rho(\mathbf{v}\cdot\mathbf{r})\mathbf{v} + p\mathbf{r}]\mathrm{d}V \\ &\quad +\int_V [\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}] : \nabla\mathbf{r}\mathrm{d}V.\end{aligned} \tag{8.6}$$

[1] Sometimes called the virial.

Where the notation $\mathbf{A} : \mathbf{B}$ denotes the inner product between two tensors:

$$\mathbf{A} : \mathbf{B} = \sum_{i=1}^{3}\sum_{k=1}^{3} A_{ik}B_{ki} = \text{Trace}\left[\begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{yx} & A_{yy} & A_{yz} \\ A_{zx} & A_{zy} & A_{zz} \end{pmatrix}\begin{pmatrix} B_{xx} & B_{xy} & B_{xz} \\ B_{yx} & B_{yy} & B_{yz} \\ B_{zx} & B_{zy} & B_{zz} \end{pmatrix}\right]. \tag{8.7}$$

Once again, we can drop the total divergence terms, since they vanish due to Gauss' theorem, to find

$$\frac{\mathrm{d}^2 I}{\mathrm{d}t^2} = -\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}V + \int_V [\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}] : \nabla\mathbf{r}\mathrm{d}V. \tag{8.8}$$

The second term in the RHS is simpler than it looks, because

$$\nabla\mathbf{r} = \begin{pmatrix} \partial_x x & \partial_x y & \partial_x z \\ \partial_y x & \partial_y y & \partial_y z \\ \partial_z x & \partial_z y & \partial_z z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}. \tag{8.9}$$

So everything simplifies to

$$\int_V [\rho\mathbf{v}\otimes\mathbf{v} + p\mathbf{I}] : \nabla\mathbf{r}\mathrm{d}V = \int_V \rho v^2 \mathrm{d}V + 3\int_V p\mathrm{d}V =: 2T + 3\Pi, \tag{8.10}$$

where T is the kinetic energy of the gas and Π is the integrated pressure.

The first term in the RHS of Equation (8.8) requires a little more work. First, we need to compute an expression for the gravitational potential in terms of the mass density ρ. To do this, we remember that the gravitational potential generated by a distribution of point masses with mass m_i at locations $\mathbf{r}_i$ is given by

$$\Psi(\mathbf{r}) = -\sum_{i=1}^{N}\frac{Gm_i}{|\mathbf{r}-\mathbf{r}_i|}. \tag{8.11}$$

This can be generalized to the case of a mass density distribution by converting the sum into an integral

$$\Psi(\mathbf{r}) = -\int_V \frac{G\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\mathrm{d}\mathbf{r}'. \tag{8.12}$$

Inserting this into the first term in the RHS of Equation (8.8) yields

$$\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}\mathbf{r} = -\int_V \mathrm{d}\mathbf{r}\rho(\mathbf{r})\nabla_{\mathbf{r}}\int_V \mathrm{d}\mathbf{r}'\frac{G\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\cdot\mathbf{r}, \tag{8.13}$$

where we have used the notation $\nabla_{\mathbf{r}}$ to remind ourselves that the derivatives are computed with respect to the unprimed position $\mathbf{r}$. In particular, note that $\rho(\mathbf{r}')$ in the integral is not affected by the derivative, since it is a function of $\mathbf{r}'$ and not of $\mathbf{r}$. The only term that we need to differentiate is the denominator:

$$\begin{aligned} &\nabla_{\mathbf{r}}\frac{1}{|\mathbf{r}-\mathbf{r}'|} \\ &= \nabla_{\mathbf{r}}\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}} \\ &= \frac{-1}{[(x-x')^2+(y-y')^2+(z-z')^2]^{3/2}} \\ &\qquad \times \begin{pmatrix} x-x' \\ y-y' \\ z-z' \end{pmatrix} \\ &= -\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}. \end{aligned} \tag{8.14}$$

Inserting the last expression into Equation (8.13), we obtain

$$
\begin{aligned}
&\int_V \rho \nabla\Psi \cdot \mathbf{r} \mathrm{d}\mathbf{r} \\
&= \int_V \mathrm{d}\mathbf{r} \int_V \mathrm{d}\mathbf{r}' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3}(\mathbf{r}-\mathbf{r}')\cdot\mathbf{r} \\
&= \int_V \mathrm{d}\mathbf{r} \int_V \mathrm{d}\mathbf{r}' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3}(\mathbf{r}-\mathbf{r}')\cdot(\mathbf{r}-\mathbf{r}') \\
&\qquad + \int_V \mathrm{d}\mathbf{r} \int_V \mathrm{d}\mathbf{r}' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3}(\mathbf{r}-\mathbf{r}')\cdot\mathbf{r}' \\
&= \int_V \mathrm{d}\mathbf{r} \int_V \mathrm{d}\mathbf{r}' \frac{G\rho(\mathbf{r}')\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|} \\
&\quad + \int_V \mathrm{d}\mathbf{r}'\rho(\mathbf{r}') \int_V \mathrm{d}\mathbf{r} \frac{G\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|^3}(\mathbf{r}-\mathbf{r}')\cdot\mathbf{r}' \\
&= -\int_V \mathrm{d}\mathbf{r}\Psi(\mathbf{r})\rho(\mathbf{r}) - \int_V \rho(\mathbf{r}')\nabla_{\mathbf{r}'}\Psi(\mathbf{r}')\cdot\mathbf{r}'\mathrm{d}\mathbf{r}'.
\end{aligned}
\tag{8.15}
$$

We have used the fact that (note the difference in sign compared to Equation (8.14))

$$
\nabla_{\mathbf{r}'}\frac{1}{|\mathbf{r}-\mathbf{r}'|} = \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}. \tag{8.16}
$$

The last term on the RHS of Equation (8.15) is identical to the LHS, so we find

$$
-\int_V \rho\nabla\Psi\cdot\mathbf{r}\mathrm{d}\mathbf{r} = \frac{1}{2}\int_V \mathrm{d}\mathbf{r}\Psi(\mathbf{r})\rho(\mathbf{r}) = \Omega, \tag{8.17}
$$

Ω being the gravitational potential energy.

For a **stationary configuration** $\mathrm{d}^2I/\mathrm{d}t^2 = 0$, so Equation (8.8) yields the *virial theorem*:

$$
\boxed{\Omega + 2T + 3\Pi = 0.} \tag{8.18}
$$

For an ideal gas $p = (\Gamma-1)\rho\varepsilon$, so

$$
\Pi = \int_V p\mathrm{d}V = (\Gamma-1)\int_V \rho\varepsilon\mathrm{d}V = (\Gamma-1)U, \tag{8.20}
$$

where U is the total internal energy. In this case, the virial theorem reads

$$
\boxed{\Omega + 2T + 3(\Gamma-1)U = 0.} \tag{8.21}
$$

8.2 Hydrostatic equilibria

The virial theorem has an important consequence for the stability of hydrostatic equilibria. Consider a cloud of gas or star in hydrostatic equilibrium, that is with no large scale motion (no rotation). Then the virial theorem states that

$$
\Omega + 3(\Gamma-1)U = 0. \tag{8.23}
$$

On the other hand, the energy of the cloud is

$$
E = \Omega + U. \tag{8.24}
$$

The energy must be negative for the system to be stable $E \leq 0$. This implies

$$
U \leq -\Omega = 3(\Gamma-1)U \Rightarrow [-1+3(\Gamma-1)]U \geq 0, \tag{8.25}
$$

since $U \geq 0$, this requires

$$
\Gamma \geq 4/3. \tag{8.26}
$$

Monoatomic gas has $\Gamma = 5/3$, so regular stars are stable. White dwarfs close to their maximum mass and very hot stars that are radiation pressure dominated have $\Gamma \approx 4/3$, so they are unstable. They explode as Type Ia supernovae and pair-instability supernova, respectively.

A surprising consequence of the virial theorem is that self-gravitating systems have *negative specific heat*. Consider a self-gravitating cloud (a protostar) emitting infrared radiation. As its energy E becomes more negative its gravitational binding energy $-\Omega$ must increase, because $U \geq 0$. We can use Equation (8.23) and Equation (8.24) to solve for U in terms of E:

$$U = -\frac{E}{3\Gamma - 4}. \tag{8.27}$$

If $\Gamma > 4/3$, Equation (8.27) predicts that U will increase as E becomes more negative. Meaning that the system heats up as it loses energy! Eventually the cloud contracts to the point that nuclear fusion starts occurring and a star is born. Nuclear burning *prevents* the star from heating up further.

8.3 Application to star formation

Consider a spherical cloud of radius R. Let the typical density be $\bar{\rho}$. Then the gravitational potential energy is (neglecting factors of order unity)

$$-\Omega = G\int_0^R 4\pi r\rho(r)\mathcal{M}(r)\mathrm{d}r \approx (4\pi)^2 G\bar{\rho}^2 R^5, \tag{8.28}$$

where

$$\mathcal{M}(r) = \int_0^r 4\pi r^2\rho(r)\mathrm{d}r. \tag{8.29}$$

For an ideal gas, the pressure is given by

$$p = \frac{k_B T}{A m_p}\rho, \tag{8.30}$$

where A is the atomic mass unit of the gas in units of the proton mass m_p, k_B is the Boltzmann constant and T is the temperature. We can assume T to be constant. This is a good approximation for optically thin clouds which can efficiently cool by emitting radiation. With this approximation

$$\Pi = \int_0^R 4\pi r^2 p(r)\mathrm{d}r \approx \frac{4}{3}\pi c_s^2\bar{\rho}R^3, \tag{8.31}$$

where

$$c_s^2 = \frac{k_B T}{A m_p} \tag{8.32}$$

is a constant. As we will see later, c_s turns out to be the sound speed of the gas. The condition for collapse of the cloud is that $3\Pi < |\Omega|$, which implies that collapse will occur for a cloud with

$$R > R_J, \qquad R_J^2 = \frac{c_s^2}{4\pi\bar{\rho}G}. \tag{8.33}$$

This is called the *Jeans length*. Typical molecular clouds have $R \gg R_J$. So instead of collapsing to an object with mass equal to the mass of the cloud, they fragment into smaller stars. The mass of these protostars can be estimated as the mass contained within one Jeans radius, the so-called *Jeans mass*[2]:

$$M_J = \frac{4\pi}{3}\bar{\rho}R_J^3 = \frac{c_s^3}{3(4\pi\bar{\rho})^{1/2}G^{3/2}} = \frac{k_B^{3/2}}{3(4\pi)^{1/2}m_p^2 G^{3/2}}\frac{T^{3/2}}{A^2 n^{1/2}}. \tag{8.34}$$

[2]Note that this definition is slightly different from the one given in C. Clarke and B. Carswell [1]; however, this difference is inconsequential given the level of approximation of these arguments.

In CGS units the Jeans mass is

$$M_J \approx 1.59 M_\odot \left(\frac{T}{\mathrm{K}}\right)^{3/2} \left(\frac{\mathrm{cm}^{-3}}{n}\right)^{1/2} \frac{1}{A^2}. \tag{8.35}$$

For a typical molecular cloud with $T = 10\,\mathrm{K}$, $n = 100\mathrm{cm}^{-3}$, and $A = 2$ (for molecular hydrogen) the Jeans mass is $M_J \approx 1.25 M_\odot$, giving a reasonable estimate for the typical mass of a fragment of the cloud. Note however that the Jeans mass decreases with increasing n, so, if the gas can efficiently cool, the cloud might undergo further fragmentation forming clusters of small stars.

While the arguments we have given here give an order of magnitude estimate for the mass of protostars, star formation is strongly influenced by a number of factors that we have neglected. These include turbulence and magnetic fields. Moreover, star formation strongly depends on the ability of the gas to cool to low temperature, as shown by Equation (8.35).

8.4 In-class worksheet

For a magnetic star the virial theorem becomes:

$$\Omega + 2T + 3\Pi + \mathfrak{M} = 0, \tag{8.36}$$

where the new term is the magnetic energy of the star, which is (in CGS units)

$$\mathfrak{M} = \frac{1}{8\pi} \int B^2 \mathrm{d}V, \tag{8.37}$$

1. Find the condition for stability of a nonrotating star in terms of Γ and $\mathfrak{M}$.
2. What is the maximum magnetic field strength for a Solar type star ($R = 7 \cdot 10^6$ km, $M = 2 \cdot 10^{33}$g)?
3. What is the maximum magnetic field strength for a neutron star ($R = 12$ km, $M = 3 \cdot 10^{33}$g)?

8.5 Exercises

Exercise 8.1 (Self-gravitating slab). An infinite, static slab of incompressible fluid with uniform density ρ_0 fills the region $|z| < a$ and is held together by its own gravity.

1. Compute the gravitational acceleration inside and outside the slab, and the pressure profile inside it.
2. Model the disk of a galaxy as such a slab, with $\rho_0 = 10^{-21}\,\mathrm{g\,cm^{-3}}$ and $a = 10^{20}$ cm. A star starts at rest at $z = a$. How fast does it cross the mid-plane, and what is the period of its vertical oscillation?

Adapted from C. Clarke and B. Carswell [1], Problem 8.

Exercise 8.2 (Mass of a molecular cloud). A self-gravitating cloud of molecular hydrogen $\left(p = nk_BT = \frac{2}{5}\rho\varepsilon\right)$ is measured to have radius $R = 0.1$ pc and temperature $T = 10$K. *Estimate* the mass of the cloud.

Exercise 8.3 (Temperature of the solar core). Estimate the temperature at the centre of the Sun, assuming that density and pressure change over a length scale comparable to the solar radius. Make the estimate twice: first assuming that the Sun is supported by gas pressure, then assuming that it is supported by radiation pressure. Which assumption is more plausible, and why?
Adapted from C. Clarke and B. Carswell [1], Problem 11.

Exercise 8.4 (Numerical solution of the Lane–Emden equation). Write a code to compute numerically the solution of the Lane–Emden equation for a given n.

1. Validate your code by comparing numerical and analytic solution for $n = 0, 1, 5$.
2. Plot the solution for $n = 1.5$ and find the radius of the polytrope, i.e., the value ξ_r at which $\theta(\xi_r) = 0$.

Hint: use the `solve_ivp` function from `scipy.integrate`.

8.6 Further reading

- C. Clarke and B. Carswell [1], Section 3.6 (the virial theorem) and Chapter 5 (hydrostatic equilibrium, polytropes, and the Lane–Emden equation).

II Waves, Shocks, and Outflows

9 Sound Waves and the Doppler Effect . 35
9.1 Oscillation Equation 35
9.2 Wave Energy and Momentum 38
9.3 In-class worksheets 41
9.4 Further reading 41

10 Waves in Stratified Atmospheres . . . 43
10.1 Waves in Isothermal Atmospheres 43
10.2 In-class worksheet 46
10.3 Exercises 46
10.4 Further reading 47

11 Shock Waves 49

12 Radiative Shocks 51

13 Blast Waves 53

14 Stellar Winds and Spherical Accretion 55

9. Sound Waves and the Doppler Effect

9.1 Oscillation Equation

The equations of fluid mechanics are nonlinear and only few analytic solutions are known. However, significant insight into astrophysical systems can be obtained by considering small perturbations around a background flow. Typically the background is chosen to be a hydrostatic solution, e.g., the density profile of an atmosphere. However, this method can be applied to more general situations. This is the key working principle of *astroseismology*. Sound waves are also one of the key tools to learn about the early Universe, as they generate the baryonic-acoustics oscillations seen in the CMB.

To set the stage, consider a gas flow with density, pressure, and velocity

$$\rho_0(t;\mathcal{V}), \quad p_0(t;\mathcal{V}), \quad \mathbf{v}_0(t;\mathcal{V}). \tag{9.1}$$

We describe its perturbation in terms of a small displacement vector $\boldsymbol{\xi}(t;\mathcal{V})$ describing the offset of each fluid element from its background state.

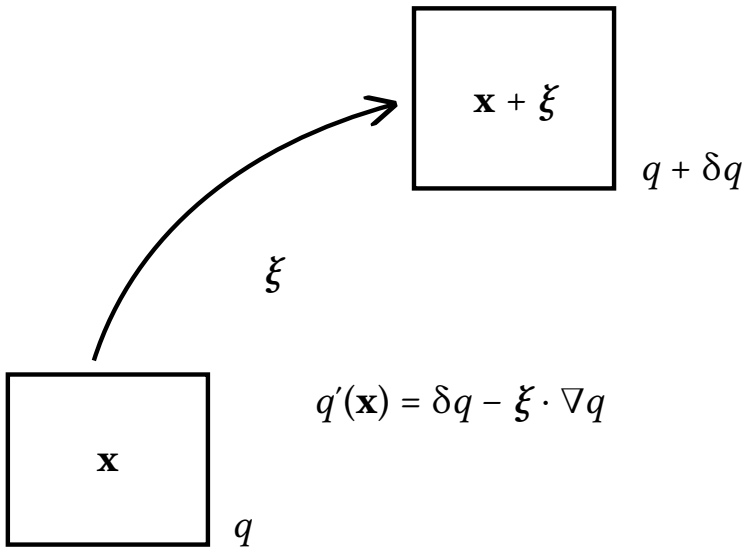


Figure 9.1: Lagrangian displacement of a fluid element from position $\mathbf{x}$ to position $\mathbf{x}+\boldsymbol{\xi}$.

The density, pressure, and velocity *of each fluid element* change according to

$$\rho(t;\mathcal{V}) = \rho_0(t;\mathcal{V}) + \delta\rho(t;\mathcal{V}), \quad p(t;\mathcal{V}) = p_0(t;\mathcal{V}) + \delta p(t;\mathcal{V}), \quad \mathbf{v}(t;\mathcal{V}) = \mathbf{v}_0(t;\mathcal{V}) + \frac{\mathrm{d}\boldsymbol{\xi}(t;\mathcal{V})}{\mathrm{d}t} \tag{9.2}$$

The quantities with the δ in front are the Lagrangian perturbations[3]. The Eulerian perturbation of a quantity is given by

$$q'(t,\mathbf{x}) = q_0(t,\mathbf{x}-\boldsymbol{\xi}) + \delta q(t,\mathbf{x}) - q_0(t,\mathbf{x}) = \delta q(t,\mathbf{x}) - \boldsymbol{\xi}\cdot\nabla q_0. \tag{9.3}$$

The difference between Lagrangian and Eulerian perturbations, is that the Lagrangian perturbations are specific to individual fluid elements, while the Eulerian perturbations record the difference in the flow at

[3] I use the common convention of denoting with δq the Lagrangian perturbations of a quantity q and with q' the Eulerian perturbations. The same convention is used in R. M. Kulsrud [2], but C. Clarke and B. Carswell [1] use Δq for Lagrangian perturbation and δq for Eulerian perturbation.

a fixed position. This distinction turns out to be crucial when perturbing non-homogeneous backgrounds, for which $\nabla q_0 \neq 0$, as we will see later.

For the moment, we consider an homogeneous medium in which ρ_0, p_0, and $\mathbf{v}_0 = 0$ are constant. We perturb it by displacing the fluid elements by a small amount $\boldsymbol{\xi}$ and let it evolve freely. How do the perturbations evolve? To find out, we need to insert the *Lagrangian* perturbations into the *Lagrangian* equations.

The continuity equation reads

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\rho\nabla\cdot\mathbf{v} \to \frac{\mathrm{d}(\rho_0+\delta\rho)}{\mathrm{d}t} = -(\rho_0+\delta\rho)\nabla\cdot\left(\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t}\right). \tag{9.4}$$

Dropping quantities that are quadratic or higher order in the perturbation, we obtain

$$\frac{\mathrm{d}\delta\rho}{\mathrm{d}t} = -\rho_0\nabla\cdot\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} \to \boxed{\delta\rho = -\rho_0\nabla\cdot\boldsymbol{\xi}} \quad (9.5) \quad (9.6)\,.$$

To find the perturbed momentum equation, first note that

$$\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} = \frac{\partial\boldsymbol{\xi}}{\partial t} + (\mathbf{v}\cdot\nabla)\boldsymbol{\xi} = \frac{\partial\boldsymbol{\xi}}{\partial t} + (\mathbf{v}_0\cdot\nabla)\boldsymbol{\xi} + \left(\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t}\cdot\nabla\right)\boldsymbol{\xi} = \frac{\partial\boldsymbol{\xi}}{\partial t} + o(|\boldsymbol{\xi}|), \tag{9.7}$$

so

$$\mathbf{v} = \frac{\partial\boldsymbol{\xi}}{\partial t}, \qquad \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\partial\mathbf{v}}{\partial t} + (\mathbf{v}_0\cdot\nabla)\mathbf{v} + \left(\frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t}\cdot\nabla\right)\mathbf{v} = \frac{\partial^2\boldsymbol{\xi}}{\partial t^2} + o(|\boldsymbol{\xi}|). \tag{9.8}$$

The momentum equation becomes

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p \to \frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\frac{1}{\rho_0}\nabla p = -\frac{1}{\rho_0}\nabla\delta p. \tag{9.9}$$

We will consider only the case of adiabatic perturbations, for which

$$p = p(\rho). \tag{9.10}$$

In other words, we are taking the pressure at constant entropy. For example, for an ideal-gas, we have already seen that for adiabatic processes

$$p = (\Gamma-1)\rho\varepsilon \to p = K\rho^\Gamma. \tag{9.11}$$

with $K = (\Gamma-1)\rho^{1-\Gamma}\varepsilon$.

In this case, we can use the chain rule to rewrite $\nabla\delta p$ in terms of $\nabla\delta\rho$:

$$\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\frac{1}{\rho_0}\frac{\mathrm{d}p}{\mathrm{d}\rho}\nabla\delta\rho = \frac{\mathrm{d}p}{\mathrm{d}\rho}\nabla\nabla\cdot\boldsymbol{\xi}. \tag{9.12}$$

Defining the *sound speed* $c_s = \sqrt{\mathrm{d}p/\mathrm{d}\rho}$, we obtain the **oscillation equation**:

$$\boxed{\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} - c_s^2\nabla\nabla\cdot\boldsymbol{\xi} = 0.} \quad (9.13)\ (9.14)$$

In general, the sound speed needs to be evaluated at constant entropy, that is

$$\boxed{c_s = \sqrt{\left(\frac{\partial p}{\partial\rho}\right)_s}\,.} \quad (9.15)\ (9.16)$$

Example

For an ideal-gas, with $p = (\Gamma - 1)\rho\varepsilon = \frac{\rho}{\mu} k_B T$, the pressure at constant entropy is

$$c_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \Gamma K \rho^{\Gamma - 1} = (\Gamma - 1)\Gamma\varepsilon = \frac{\Gamma k_B T}{\mu}. \tag{9.17}$$

The oscillation Equation (9.13) is the fundamental equation to describe sound waves. Taking the divergence on both sides and using Equation (9.5), we find that the density perturbation $\delta\rho$ satisfies the wave equation:

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \nabla^2 \delta\rho = 0. \tag{9.18}$$

To solve the oscillation equation, we consider the ansatz

$$\boldsymbol{\xi} = \boldsymbol{\xi}_1 \exp[\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)], \tag{9.19}$$

which describes a plane sound wave propagating in the direction $\mathbf{k}$ and having frequency ω. $\boldsymbol{\xi}_1$ is the *polarization vector*, describing the way in which fluid elements oscillates as the wave propagates. Inserting this ansatz into Equation (9.13), we find

$$\omega^2 \boldsymbol{\xi}_1 = c_s^2 \mathbf{k}(\mathbf{k}\cdot\boldsymbol{\xi}_1). \tag{9.20}$$

This means that plane wave solutions to Equation (9.13) exist only for specific choices of $\boldsymbol{\xi}_1, \omega$ and $\mathbf{k}$. This is an example of dispersion relation. We see that the RHS is parallel to $\mathbf{k}$, so solutions can exist only if $\boldsymbol{\xi}_1 \parallel \mathbf{k}$. This means that sound waves are longitudinal. This is opposite from electromagnetic waves in vacuum, in which the electric and magnetic fields oscillate in the plane perpendicular to the direction of propagation. Setting $\boldsymbol{\xi}_1 = \xi_1 \hat{\mathbf{k}}$, we find

$$\omega^2 = c_s^2 k^2. \tag{9.21}$$

This means that both phase and group velocity for sound waves is given by c_s

$$v_{\text{phase}} = \frac{\omega}{k} = c_s, \qquad v_{\text{group}} = \frac{\mathrm{d}\omega}{\mathrm{d}k} = c_s. \tag{9.22}$$

This justifies the name *sound speed* for c_s.

What happens to fluid elements when a sound wave passes through them? They are displaced along the direction of the wave. This causes a compression of the fluid. From Equation (9.5), we find:

$$\delta\rho = -\rho_0 (\mathbf{k}\cdot\boldsymbol{\xi}_1) e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}. \tag{9.23}$$

Example

Consider a sound wave propagating along z with wavelength $\lambda = 1$. Then $\mathbf{k} = k\begin{pmatrix}0 & 0 & 1\end{pmatrix}^T$ and $\omega = c_s$. The only non-zero component of $\boldsymbol{\xi}$ is the z-component

$$\xi_z = \xi_1 e^{\mathrm{i}(z - c_s t)} = A\cos[k(z - c_s t)] + B\sin[k(z - c_s t)] \Leftrightarrow \xi_1 = A - \mathrm{i}B \tag{9.24}$$

The density perturbation is given by

$$\frac{\delta\rho}{\rho_0} = -\frac{\partial \xi_z}{\partial z} = k\{A\sin[k(z - c_s t)] - B\cos[k(z - c_s t)]\}. \tag{9.25}$$

9.2 Wave Energy and Momentum

Sound waves also transport energy and momentum. The deposition of energy by breaking waves is thought to play an important role in phenomena such as core-collapse supernovae, stellar eruptions, and the evolution of giant molecular clouds.

i Going further: Normal Modes

Before we can compute the energy carried by waves, we need to develop some general perturbation theory. In general, the oscillation equation takes the form

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \hat{\mathbf{F}}\boldsymbol{\xi}, \tag{9.26}$$

where $\hat{\mathbf{F}}$ is a differential operator. For example, for sound waves in a homogeneous medium $\hat{\mathbf{F}}\boldsymbol{\xi} = \nabla\nabla\cdot\boldsymbol{\xi}$. We can find a dispersion relation by looking for solutions such that

$$\boldsymbol{\xi}(t,\mathbf{x}) = \boldsymbol{\eta}(\mathbf{x})e^{-\mathrm{i}\omega t}. \tag{9.27}$$

Inserting this ansatz in Equation (9.26), we find an eigenvalue problem

$$-\omega^2\rho_0\boldsymbol{\eta}(\mathbf{x}) = \hat{\mathbf{F}}\boldsymbol{\eta}(\mathbf{x}). \tag{9.28}$$

In most applications, $\hat{\mathbf{F}}$ is self-adjoint:

$$\int \boldsymbol{\zeta}(\mathbf{x})\cdot\hat{\mathbf{F}}\boldsymbol{\xi}(\mathbf{x})\rho_0(\mathbf{x})\mathrm{d}^3\mathbf{x} = \int \boldsymbol{\xi}(\mathbf{x})\cdot\hat{\mathbf{F}}\boldsymbol{\zeta}(\mathbf{x})\rho_0(\mathbf{x})\mathrm{d}^3\mathbf{x}. \tag{9.29}$$

The general theory of linear operators then implies that Equation (9.28) has an infinite set of eigenvalues $\{\omega_n\}$ and eigenfunctions $\{\boldsymbol{\eta}_n\}$ and that any perturbation (not just plane waves) can be written as an infinite superposition of elementary solutions:

$$\boldsymbol{\xi}(t,\mathbf{x}) = \sum_n \boldsymbol{\eta}_n(\mathbf{x})e^{-i\omega_n t}. \tag{9.30}$$

The quantities $\boldsymbol{\eta}_n(\mathbf{x})$ are also called the *normal modes of oscillations*. Note that the normal modes need not be countable. For example, for sound waves in a homogeneous infinite medium all positive sound frequencies ω are allowed. In this case, Equation (9.30) takes the form of an integral. For plane waves in a homogeneous medium, Equation (9.30) is just the Fourier transform:

$$\boldsymbol{\xi}(t,\mathbf{x}) = \int \mathrm{d}^3\mathbf{k}\boldsymbol{\eta}(\mathbf{k})e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}e^{-\mathrm{i}c_s kt}. \tag{9.31}$$

However, when we consider sound waves in a finite medium, for example a star of a given radius, the requirement that standing wave solutions have a wavelength that is an integer fraction of the dimension of the star, makes the frequencies $\{\omega_n\}$ quantized.

Example

Consider a rigid, cylindrical pipe of length L filled with gas at constant density and temperature. The ends of the pipe are sealed. The displacement equation for sound waves propagating in the pipe is

$$\begin{cases} \partial_t^2\xi(t,z) = c_s^2\partial_z^2\xi(t,z), 0<z<L, \\ \xi(t,0) = \xi(t,L) = 0. \end{cases} \tag{9.32}$$

If we look for solutions in the form $\xi(t,z) = \eta(z)e^{\mathrm{i}\omega t}$, we find the eigenvalue problem

$$\begin{cases} \partial_z^2\eta(z) + \left(\frac{\omega}{c_s}\right)^2\eta(z) = 0, 0<z<L \\ \eta(0) = \eta(L) = 0. \end{cases} \tag{9.33}$$

This equation is just the equation for a harmonic oscillator. The general solution is

$$\eta(z) = A\cos(\kappa z) + B\sin(\kappa z), \qquad \kappa = \frac{\omega}{c_s}. \tag{9.34}$$

However, not all solutions satisfy the boundary conditions. To satisfy the boundary conditions, we require

$$A = 0, \qquad \kappa = \frac{n\pi}{L} \Rightarrow \omega = c_s\left(\frac{n\pi}{L}\right), \quad n = 1,2,\ldots \tag{9.35}$$

The first few normal modes are plotted in Figure 9.2.

i Going further: Work Integral

We might be tempted to define the energy density of a wave in the general form $\boldsymbol{\xi}(t, \mathbf{x}) = \boldsymbol{\eta}(\mathbf{x})e^{\mathrm{i}\omega t}$ as

$$K_{\text{wave}} = \frac{1}{2}\rho_0 \left\langle \left| \frac{\partial \boldsymbol{\xi}}{\partial t} \right|^2 \right\rangle = \frac{1}{2}\rho_0 |\boldsymbol{\eta}|^2 \langle |\omega \sin(\omega t)|^2 \rangle = \frac{\omega}{4}\rho_0 |\boldsymbol{\eta}|^2, \tag{9.37}$$

where $\langle \cdot \rangle$ denotes averaging over a period $T = 2\pi/\omega$ of the wave. Equation (9.37) is the kinetic energy associated with the wave. However, it turns out that the kinetic energy is just one portion of the wave energy.

The energy of the wave is related to ξ^2, so it cannot be computed directly from linear theory. However, we can use a trick:

1. adiabatically perturb the system to create a wave with an external force;
2. compute the work done by the external force on the system.

To achieve this we add a source term to the oscillation equation:

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \hat{\mathbf{F}}\boldsymbol{\xi} + \mathbf{F}_{\text{ext}}(t). \tag{9.38}$$

Suppose we want to excite the mode $\boldsymbol{\xi}(t, \mathbf{x}) = \boldsymbol{\eta}(\mathbf{x})e^{-\mathrm{i}\omega t}$, where $\boldsymbol{\eta}$ is such that $\hat{\mathbf{F}}\boldsymbol{\eta} = -\rho_0\omega^2\boldsymbol{\eta}$. Then we can take $\mathbf{F}_{\text{ext}}(t) = F_0 e^{\gamma t}\boldsymbol{\eta}(\mathbf{x})$, where F_0 is chosen so that for $t \lesssim 0$, $\boldsymbol{\xi}(t, \mathbf{x}) = e^{\gamma t}\boldsymbol{\eta}(\mathbf{x})$. Inserting this into the oscillation equation yields

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \rho_0 \gamma^2 e^{\gamma t}\boldsymbol{\eta} = e^{\gamma t}\hat{\mathbf{F}}\boldsymbol{\eta} + F_0 e^{\gamma t}\boldsymbol{\eta} = (-\rho_0\omega^2 + F_0)e^{\gamma t}\boldsymbol{\eta}. \tag{9.39}$$

We can solve for F_0 to find that

$$F_0 = \rho_0(\omega^2 + \gamma^2). \tag{9.40}$$

The work done by the external force per unit volume is given by the *work integral*

$$W = \int_{-\infty}^{0} \mathrm{d}t \frac{\partial \boldsymbol{\xi}}{\partial t} \cdot \mathbf{F}_{\text{ext}} = \int_{-\infty}^{0} \mathrm{d}t \rho_0 \gamma(\gamma^2 + \omega^2) e^{2\gamma t}\boldsymbol{\eta} \cdot \boldsymbol{\eta} = \rho_0 |\boldsymbol{\eta}|^2 \int_{-\infty}^{0} \mathrm{d}t \gamma(\gamma^2 + \omega^2) e^{2\gamma t}, \tag{9.41}$$

where we have used the fact that $\boldsymbol{\eta}$ is time independent. The last integral can be computed analytically, so the previous expression yields

$$W = \rho_0 |\boldsymbol{\eta}|^2 \left(\frac{\gamma^2}{2} + \frac{\omega^2}{2} \right). \tag{9.42}$$

The energy of the wave is obtained in the limit in which $\gamma \downarrow 0$:

$$E_{\text{wave}} = \frac{\omega}{2}\rho_0 |\boldsymbol{\eta}|^2. \tag{9.43}$$

Note that $E_{\text{wave}} = 2K_{\text{wave}}$, meaning that half of the wave energy is potential energy. For example, for a sound wave with $\boldsymbol{\eta} = \mathbf{A}\cos(\mathbf{k} \cdot \mathbf{x})$, the mean energy density (averaging over one wave-length) is

$$E_{\text{wave}} = \frac{\omega}{4}\rho_0 A^2. \tag{9.45}$$

It is possible to compute the linear momentum $\mathbf{P}_{\text{wave}}$ carried by waves in a similar way. One finds

$$\mathbf{P}_{\text{wave}} = \frac{\mathbf{k}}{\omega} E_{\text{wave}}. \tag{9.46}$$

9.3 In-class worksheets

9.3.1 Sound waves

Consider an ideal gas with

$$p = (\Gamma - 1)\rho\varepsilon = \frac{\rho}{\mu} k_B T. \tag{9.48}$$

What is the sound speed c_s as a function of ρ and T?
Show that

$$\boldsymbol{\xi} = f(z - c_s t)\hat{\mathbf{z}} \tag{9.49}$$

is a solution of

$$\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} - c_s^2 \nabla\nabla \cdot \boldsymbol{\xi} = 0. \tag{9.50}$$

Sketch the solution for different times if $f(x) = e^{-x^2/2}$.

9.3.2 The Doppler effect

Consider an initially homogeneous flow with

$$\rho = \rho_0 = \text{const}, \qquad p = p_0 = \text{const}, \qquad \mathbf{v} = \mathbf{v}_0 = \text{const}\ . \tag{9.51}$$

Introduce a small perturbation in the form of a Lagrangian displacement $\boldsymbol{\xi}$.

1. Show that the Lagrangian density perturbation is given by
$$\delta\rho = -\rho_0 \nabla \cdot \boldsymbol{\xi}. \tag{9.52}$$
2. Use the results from part 1) to show that the oscillation equation reads
$$\left[\frac{\partial}{\partial t} + (\mathbf{v}_0 \cdot \nabla)\right]^2 \boldsymbol{\xi} = -c_s^2 \nabla\nabla \cdot \boldsymbol{\xi}. \tag{9.53}$$
3. Introduce the ansatz $\boldsymbol{\xi} = \boldsymbol{\eta} e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}$ and show that the dispersion relation is
$$\omega_\pm = \mathbf{v}_0 \cdot \mathbf{k} \pm c_s k. \tag{9.54}$$
What is the physical meaning of the $\pm$ solutions?
4. Consider a sound wave propagating in the direction of the flow. Let $\omega_0 = c_s k$ be the frequency of sound in the fluid frame. Show that the frequency of sound in the lab frame is
$$\omega = \omega_0 \left(1 + \frac{v}{c_s} \cos\theta\right), \tag{9.55}$$
where $\cos\theta$ is the angle between the direction in which sound propagates and $\mathbf{v}_0$.

9.4 Further reading

- C. Clarke and B. Carswell [1], Sections 6.1 and 6.3.

10. Waves in Stratified Atmospheres

10.1 Waves in Isothermal Atmospheres

Consider a stratified, static, isothermal atmosphere. We assume constant acceleration g in the negative z direction. The hydrostatic equilibrium equations are

$$\partial_z p_0 = -\rho_0 g, \qquad p = a^2 \rho, \tag{10.1}$$

where $a = \sqrt{\frac{kT}{\mu}}$ is the isothermal sound speed.

Combining these two we find

$$\partial_z \rho_0 = -\frac{g}{a^2}\rho_0 \Rightarrow \rho_0 = \tilde{\rho} e^{-z/H}. \tag{10.2}$$

$H = a^2/g$ is called the scale-high of the atmosphere and has units of a length.

We want to study the propagation of waves in this atmosphere. The continuity equation still gives

$$\delta\rho = -\rho_0 \nabla \cdot \boldsymbol{\xi}. \tag{10.3}$$

and the velocity perturbation is

$$\mathbf{v}' = \delta\mathbf{v} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{v}_0 = \delta\mathbf{v} = \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}t} = \frac{\partial\boldsymbol{\xi}}{\partial t} + (\mathbf{v}_0\cdot\nabla)\boldsymbol{\xi} = \frac{\partial\boldsymbol{\xi}}{\partial t}. \tag{10.4}$$

The sound speed is given by

$$c_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \Gamma a^2. \tag{10.5}$$

Which implies that $\delta p = \Gamma a^2 \delta\rho$ for adiabatic perturbations. However,

$$p' \overset{!}{=} \Gamma a^2 \rho', \tag{10.6}$$

instead we have

$$\delta p = \Gamma a^2 \delta\rho \Rightarrow (p' + \boldsymbol{\xi}\cdot\nabla p_0) = \Gamma a^2 \delta\rho \Rightarrow p' = \Gamma a^2 \delta\rho + g\rho_0 \xi_z. \tag{10.7}$$

The momentum equation reads

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p - g\rho\hat{\mathbf{z}}. \tag{10.8}$$

Setting $\mathbf{v} = \mathbf{v}'$, $p = p_0 + p'$, and $\rho = \rho_0 + \rho'$ yields

$$\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\nabla p_0 - g\rho_0\hat{\mathbf{z}} - \nabla p' - g\rho'\hat{\mathbf{z}} + \text{HOT}\ . \tag{10.9}$$

The first two terms on the RHS cancel out. Dropping the higher order terms we find

$$\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -\Gamma a^2\nabla\delta\rho + g\frac{\rho_0}{H}\xi_z\hat{\mathbf{z}} - g\rho_0\nabla\xi_z - g\Big(\delta\rho + \frac{\rho_0}{H}\xi_z\Big)\hat{\mathbf{z}}, \tag{10.10}$$

where we have used Equation (10.7) and

$$\rho' = \delta\rho - (\boldsymbol{\xi}\cdot\nabla)\rho_0 = \delta r + \frac{\rho_0}{H}\xi_z. \tag{10.11}$$

The second and last term on the RHS of Equation (10.10) cancel out leaving

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = -\Gamma a^2 \nabla \delta\rho - g\rho_0 \nabla \xi_z - g\delta\rho \hat{\mathbf{z}}. \tag{10.12}$$

Using the fact that

$$\begin{aligned} \nabla\delta\rho &= \nabla(-\rho_0 \nabla \cdot \boldsymbol{\xi}) \\ &= -\rho_0 \nabla\nabla \cdot \boldsymbol{\xi} - \nabla\rho_0 \nabla \cdot \boldsymbol{\xi} \\ &= -\rho_0 \nabla\nabla \cdot \boldsymbol{\xi} + \frac{\rho_0}{H} \nabla \cdot \boldsymbol{\xi}\hat{\mathbf{z}}, \end{aligned} \tag{10.13}$$

We obtain

$$\begin{aligned} \rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \Gamma a^2 \rho_0 \nabla\nabla \cdot \boldsymbol{\xi} - \rho_0 \Gamma g \nabla \cdot \boldsymbol{\xi}\hat{\mathbf{z}} \\ -g\rho_0 \nabla \xi_z + g\rho_0 \nabla \cdot \boldsymbol{\xi}\hat{\mathbf{z}}. \end{aligned} \tag{10.14}$$

This can be simplified to

$$\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = c_s^2 \left(\nabla\nabla \cdot \boldsymbol{\xi} - \frac{a^2}{c_s^2} \frac{1}{H} \nabla \xi_z \right) \tag{10.15}$$

$$-\frac{(c_s^2 - a^2)}{H} \nabla \cdot \boldsymbol{\xi}\hat{\mathbf{z}}. \tag{10.16}$$

We are going to analyze the consequences of this equation shortly. Before we do that, we show that this equation reduces to Eq. (6.31) of C. Clarke and B. Carswell [1] in the limit of vertical oscillations in the z direction. To do that, we take the divergence of Equation (10.15) to obtain

$$\begin{aligned} \frac{\partial^2 \delta\rho}{\partial t^2} &= c_s^2 \left(\nabla^2 \delta\rho + \frac{\delta\rho}{H^2} \right) + g\rho_0 \nabla^2 \xi_z \\ &\quad -\frac{(c_s^2 - a^2)}{H} \left(\partial_z \delta\rho + \frac{\delta\rho}{H} \right) \\ &= c_s^2 \nabla^2 \delta\rho - \frac{c_s^2}{H} \partial_z \delta\rho + \frac{a^2}{H} \rho_0 \nabla^2 \xi_z \\ &\quad + \frac{a^2}{H} \partial_z \delta\rho + a^2 \frac{\delta\rho}{H^2}. \end{aligned} \tag{10.17}$$

Assuming that $\delta\rho = \delta\rho(t, z)$ and $\boldsymbol{\xi} = \xi_z \hat{\mathbf{z}}$, we obtain,

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \frac{\partial^2 \delta\rho}{\partial z^2} - \frac{c_s^2}{H} \frac{\partial \delta\rho}{\partial z} = 0, \tag{10.18}$$

where we have used the fact that

$$\nabla^2 \xi_z = \partial_z^2 \xi_z = \partial_z \left(-\frac{\delta\rho}{\rho_0} \right) = -\frac{\partial_z \delta\rho}{\rho_0} - \frac{\delta\rho}{\rho_0 H}. \tag{10.19}$$

Equation (10.18) is identical to Eq. (6.31) of C. Clarke and B. Carswell [1]. It describes the propagation of density modes in a stratified atmosphere, but neglects an entirely different class of modes that have $\boldsymbol{\xi}$ in the z-direction, but propagate in the xy-plane.

As before, we look for traveling wave solutions

$$\boldsymbol{\xi} = \boldsymbol{\eta} e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x} - \omega t)}. \tag{10.20}$$

Inserting this into Equation (10.15) we obtain

$$\omega^2\boldsymbol{\eta} = c_s^2\left(\mathbf{k}\cdot\boldsymbol{\eta} + \mathrm{i}\frac{a^2}{c_s^2}\frac{1}{H}\eta_z\right)\mathbf{k} \tag{10.21}$$

$$+\mathrm{i}\frac{(c_s^2-a^2)}{H}\mathbf{k}\cdot\boldsymbol{\eta}\hat{\mathbf{z}}. \tag{10.22}$$

- In the limit of $H \gg a^2k/\omega^2$, that is, in the limit in which the wavelength of perturbations is much smaller than the scale height of the atmosphere, we recover the dispersion relation for waves in a homogeneous medium, namely $\omega^2 = c_s k^2$, as expected.
- Purely horizontal waves, with $\boldsymbol{\eta}, \mathbf{k} \perp \hat{\mathbf{z}}$ cannot satisfy the dispersion relation, unless $c_s = a$, that is for isothermal as opposed to adiabatic perturbations. Physically this means that density fluctuations become buoyant, so they acquire a $k_z \neq 0$ component in their wave vector.
- Waves with $\boldsymbol{\eta} \parallel \mathbf{k} \parallel \hat{\mathbf{z}}$ satisfy the dispersion relation simplifies to

$$\omega^2 = c_s^2\left(k^2 + \mathrm{i}\frac{1}{H}k\right) \Rightarrow k_{1,2} = -\frac{\mathrm{i}}{2H} \pm \sqrt{\left(\frac{\omega}{c_s}\right)^2 - \left(\frac{1}{2H}\right)^2}. \tag{10.23}$$

  These are the so called pressure or p-modes of the atmosphere. They are the generalization of sound waves.

  – If $H \gg \frac{c_s}{\omega}$, then the dispersion relation for p-modes can be approximated as:

$$k_{1,2} \approx -\frac{\mathrm{i}}{2H} \pm \frac{\omega}{c_s}. \tag{10.24}$$

    In this case the displacement grows exponentially with height:

$$\xi_z = \eta_z e^{\mathrm{i}(\mathbf{k}\cdot\mathbf{x}-\omega t)} = \eta_z \exp\left[\mathrm{i}\omega\left(\pm\frac{z}{c_s} - t\right)\right]\exp\left(\frac{z}{2H}\right). \tag{10.25}$$

    This means that waves propagating vertically along an isothermal atmosphere eventually become nonlinear and will experience wave-breaking. Note that the density fluctuation remains finite:

$$\delta\rho = -\rho_0\nabla\cdot\boldsymbol{\xi} = -\tilde{\rho}\eta_z\left(\frac{\mathrm{i}\omega}{c_s} + \frac{1}{2H}\right)\exp\left[\mathrm{i}\omega\left(\frac{z}{c_s} - t\right)\right]\exp\left(-\frac{z}{2H}\right). \tag{10.26}$$

  – If $H \ll \frac{c_s}{\omega}$, then the solution describes one of two global oscillation modes of the atmosphere:

$$k_{1,2} \approx -\frac{\mathrm{i}}{H}, 0 \qquad \Rightarrow \qquad [\xi_z]_{1,2} = \eta e^{\mathrm{i}\omega t}, \eta_z e^{z/H} e^{\mathrm{i}\omega t}. \tag{10.27}$$

- Let us now consider the case of perturbations with horizontal displacement. We can align the x-axis with $\boldsymbol{\eta}$, meaning that $\boldsymbol{\eta} = \eta\hat{\mathbf{x}}$. In this case, the dispersion relation becomes

$$\omega^2\eta\hat{\mathbf{x}} = c_s^2(k_x\eta)\mathbf{k} + \mathrm{i}\frac{(c_s^2-a^2)}{H}k_x\eta\hat{\mathbf{z}}. \tag{10.28}$$

  This means that there are two components of $\mathbf{k} = \begin{pmatrix} k_x & 0 & k_z \end{pmatrix}$. The x-component of Equation (10.28) is

$$\omega^2 = c_s^2 k_x^2 \Rightarrow k_x = \pm\frac{\omega}{c_s}. \tag{10.29}$$

  The z-component of Equation (10.28) is

$$k_z = -\mathrm{i}\left(1 - \frac{a^2}{c_s^2}\right)\frac{1}{H}. \tag{10.30}$$

  The corresponding displacement is

$$\boldsymbol{\xi}(t,\mathbf{x}) = \eta \exp\left[\mathrm{i}\omega\left(\pm\frac{x}{c_s} - t\right)\right] \exp\left[\left(1 - \frac{a^2}{c_s^2}\right)\frac{z}{H}\right]\hat{\mathbf{x}}. \tag{10.31}$$

Note that, as in the previous case, the density fluctuations $\delta\rho$ remain finite. These are the so called (internal) gravity-modes or g-modes of the atmosphere. In the limit in which $c_s \to a$, that is $\Gamma \to 1$ (isothermal perturbations), these type of waves reduce to horizontal sound waves.

10.2 In-class worksheet

We consider small disturbances in a plane parallel, isothermal atmosphere. We will assume z to be the vertical direction and g to be the gravitational acceleration, assumed to be constant and towards the negative z direction. Let $a = \sqrt{k_B T/\mu}$ be the isothermal sound speed.

1. Show that the static background density ρ_0 satisfies

$$\rho_0(z) = \tilde{\rho} e^{-z/H}, \tag{10.32}$$

where $H = a^2/g$ is the scale height of the atmosphere.

2. Let $\delta\rho$ and ρ' be the Lagrangian and Eulerian perturbations of the density. Show that

$$\rho' = \delta\rho + \frac{\rho_0}{H}\xi_z. \tag{10.33}$$

3. Let p' be the Eulerian pressure perturbation. Show that

$$p' = c_s^2\delta\rho + g\rho_0\xi_z, \tag{10.34}$$

where $c_s^2 = \Gamma a^2$ is the square of the sound speed and ξ_z is the z-component of the displacement.

4. Use the results from the previous two points to show that

$$\rho_0\frac{\partial^2\boldsymbol{\xi}}{\partial t^2} = -c_s^2\nabla\delta\rho - g\rho_0\nabla\xi_z - g\delta\rho\hat{\mathbf{z}}. \tag{10.35}$$

10.3 Exercises

Exercise 10.1 (Waves in a polytropic atmosphere). An isentropic atmosphere sits in a uniform gravitational field g, with pressure $p_0 = A\rho_0^{1+1/n}$ and A constant. Let x be the depth below the top of the atmosphere, and suppose that the unperturbed density is $\rho_0 = (kx)^n$ with k constant.

1. Show that hydrostatic equilibrium requires $g = Ak(1+n)$.
2. Perturb the atmosphere with a small vertical velocity u, taken positive upward, together with adiabatic perturbations $p' \ll p_0$ of the pressure and $\rho' \ll \rho_0$ of the density. Show that $p' = (gx/n)\rho'$.
3. Show that the linearized continuity and momentum equations are

$$\frac{\partial\rho'}{\partial t} - u\frac{\partial\rho_0}{\partial x} - \rho_0\frac{\partial u}{\partial x} = 0 \tag{10.36}$$

and

$$\frac{\partial u}{\partial t} + \left(\frac{\rho'}{\rho_0}\right)g - \frac{g}{n\rho_0}\frac{\partial(x\rho')}{\partial x} = 0. \tag{10.37}$$

4. For $n = 1/2$, combine these equations to show that

$$\frac{\partial^2 u}{\partial t^2} = 2gx\frac{\partial^2 u}{\partial x^2} + 3g\frac{\partial u}{\partial x}. \tag{10.38}$$

5. Look for solutions $u(t,x) = u_1(x)e^{\mathrm{i}\omega t}$ and change variable to $\tau = (2x/g)^{1/2}$. Show that Equation (10.38) becomes

$$\frac{1}{\tau}\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}(\tau u_1) + \omega^2 u_1 = 0. \tag{10.39}$$

Find the solution that stays finite at $x = 0$ and interpret it physically.

Adapted from C. Clarke and B. Carswell [1], Problem 22.

10.4 Further reading

- C. Clarke and B. Carswell [1], Section 6.2.

11. Shock Waves

> **Note**
>
> This chapter is in preparation.

12. Radiative Shocks

> **Note**
>
> This chapter is in preparation.

13. Blast Waves

> **Note**
>
> This chapter is in preparation.

14. Stellar Winds and Spherical Accretion

> **Note**
>
> This chapter is in preparation.

III Instabilities, Viscosity, Disks, and Turbulence

15 Fluid Instabilities I: Convection 59

16 Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz 61

17 Fluid Instabilities III: Thermal and Gravitational Instabilities 63

18 Diffusion and Viscous Flows 65

19 Accretion Disks I: Structure and Viscous Evolution 67

20 Accretion Disks II: Steady Disks and Spectra 69

21 Turbulence 71

15. Fluid Instabilities I: Convection

> **Note**
>
> This chapter is in preparation.

16. Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz

> **Note**
>
> This chapter is in preparation.

17. Fluid Instabilities III: Thermal and Gravitational Instabilities

> **Note**
>
> This chapter is in preparation.

18. Diffusion and Viscous Flows

> **Note**
>
> This chapter is in preparation.

19. Accretion Disks I: Structure and Viscous Evolution

> **Note**
>
> This chapter is in preparation.

20. Accretion Disks II: Steady Disks and Spectra

> **Note**
>
> This chapter is in preparation.

21. Turbulence

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

IV Plasma and Magnetohydrodynamics

22 Introduction to Plasma 75

23 The MHD Equations 77

24 Flux Freezing 79

25 Magnetic Tension and Hydromagnetic Equilibria 81

26 MHD Waves 83

27 The Magnetorotational Instability . . 85

27.1 Qualitative Picture 85
27.2 Detailed Calculation 86
27.3 In-class worksheet 89
27.4 Further reading 89

22. Introduction to Plasma

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

23. The MHD Equations

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

24. Flux Freezing

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

25. Magnetic Tension and Hydromagnetic Equilibria

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

26. MHD Waves

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

27. The Magnetorotational Instability

The magnetorotational instability (MRI) was first discovered in the context of plasma fusion reactors by E. P. Velikhov [3] and S. Chandrasekhar [4] and then rediscovered in the context of accretion disks by S. A. Balbus and J. F. Hawley [5]. This is the instability thought to be responsible for the α-viscosity in many astrophysical systems: from X-ray binaries to AGN disks. It is one of the most important, among the many instabilities operating in plasmas, so we will cover it in detail in these notes.

It was long known that a way to transport angular momentum in the disk is through magnetic stresses, that is the $B_r B_\theta / 4\pi$ term in the momentum equation. However, to explain the observed luminosities from accretion disks, the magnetic stresses must be similar in magnitude to the pressure in the disk p (the proportionality constant being the α-viscosity coefficient). Given a small component of the field in the radial direction, flux freezing can grow B_θ linearly in time. However, since B_r is small, this mechanism can transport angular momentum efficiently only if $B_\theta^2 \gg p$. Such a strong toroidal field cannot be contained in the disk. The requirement for the magnetic stresses to do the job of transporting angular momentum is that $B_r^2 \approx B_\theta^2 \approx 4\pi\rho c_s^2$. Detailed simulations have shown that MRI is a mechanism that naturally produces the required field strengths over a short timescale.

27.1 Qualitative Picture

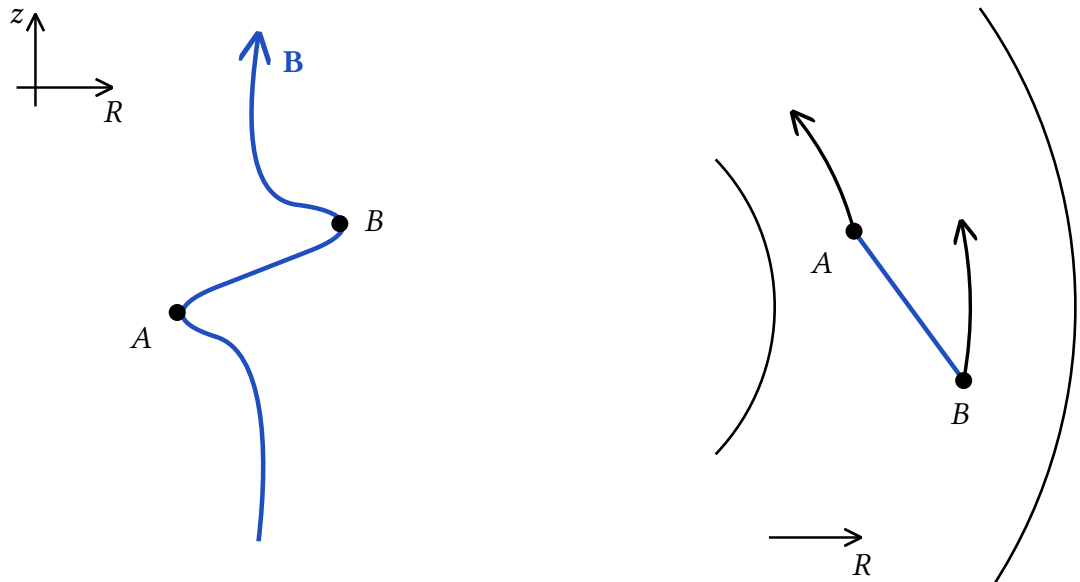


Figure 27.1: Qualitative description of the magnetorotational instability transporting angular momentum between fluid elements A and B. *Left panel:* meridional view. *Right panel:* orbital plane.

Consider an accretion disk with Keplerian angular velocity profile

$$\Omega(R) = \Omega_0 \left(\frac{R_0}{R} \right)^{3/2}, \tag{27.1}$$

initially threaded by a vertical magnetic field $B_0 \hat{\mathbf{z}}$. Let A and B be two fluid elements connected by a magnetic field line (initially vertical) and displace their positions according to $\xi_r = \epsilon \cos kz$, as shown in Figure 27.1. The fluid elements A and B immediately accelerate to match the local angular velocity.

However, as a result of the change in angular velocity, a toroidal component of the field, B_θ is generated by the stretching of the original field line.

The magnetic tension then creates an attractive force between A and B, which has the effect of transporting angular momentum from A to B. This causes A to lose angular momentum and sink further inwards, while B gains angular momentum and starts moving outwards. In turn, because the angular velocity decreases with radius, A increases its angular velocity, while B slows down, further separating the two points. This in turn enhances the magnetic tension force. The cycle we have just described is unstable, so the initial perturbation grows exponentially in time, until the system becomes nonlinear.

The mechanism we have just described involves the magnetic tension force, so it is natural to expect a characteristic growth rate of

$$\gamma \approx k v_A, \qquad v_A = \frac{B_z}{\sqrt{4\pi\rho}}. \tag{27.2}$$

In our derivation, we have assumed the field to be sufficiently weak, so that the displaced fluid elements can acquire the angular velocity of their surroundings. This should be valid as long as

$$\omega_A = k v_A \lesssim \Omega. \tag{27.3}$$

We will find that the exact condition for instability is $k v_A < \sqrt{3}\Omega$, so the maximum growth rate is achieved by taking k sufficiently large, so that $\gamma \approx k v_A \approx \Omega$. This is a very short timescale: the MRI grows over the orbital time scale! Moreover, the growth rate is independent of the field strength, as long as v_A is not so large that the required k to satisfy $k v_A < \sqrt{3}\Omega$ corresponds to perturbation with characteristic size larger than the vertical size of the disk.

27.2 Detailed Calculation

The starting point is the Euler equation, which we write in the usual form in cylindrical coordinates (R, θ, z):

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v}\right] = -\nabla\left(p + \frac{B^2}{8\pi}\right) + \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{4\pi} + \mathbf{g}\rho. \tag{27.4}$$

We consider a disk that is Keplerian and threaded by a vertical magnetic field:

$$\mathbf{v} = R\Omega\hat{\boldsymbol{\theta}} + \mathbf{u}, \qquad \mathbf{B} = B_0\hat{\mathbf{z}} + \mathbf{b}, \qquad p + \frac{B^2}{8\pi} = P_0 + P. \tag{27.5}$$

We will treat $\mathbf{u}$ and $\mathbf{b}$ as (small) Eulerian perturbations. We will only consider modes that are axisymmetric (no θ dependence) and incompressible

$$\nabla\cdot\mathbf{u} = \frac{1}{R}\frac{\partial(R u_r)}{\partial R} + \frac{\partial u_z}{\partial z} = 0. \tag{27.6}$$

To leading order in $\mathbf{u}$ (using Kepler's law)

$$\begin{aligned}(\mathbf{v}\cdot\nabla)\mathbf{v} \to (R\Omega)\hat{\boldsymbol{\theta}}\cdot\nabla\mathbf{u} + \mathbf{u}\cdot\nabla(R\Omega\hat{\boldsymbol{\theta}}) = \\ -2\Omega u_\theta\hat{\mathbf{R}} + \left(R\frac{\mathrm{d}\Omega}{\mathrm{d}R} + 2\Omega\right)u_R\hat{\boldsymbol{\theta}} = \\ \frac{1}{2}\Omega u_R\hat{\boldsymbol{\theta}} - 2\Omega u_\theta\hat{\mathbf{R}}.\end{aligned} \tag{27.7}$$

Similarly,

$$\begin{aligned}(\mathbf{B}\cdot\nabla)\mathbf{B} \to B_0\hat{\mathbf{z}}\cdot\nabla\mathbf{b} + \mathbf{b}\cdot\nabla(B_0\hat{\mathbf{z}}) = \\ B_0\frac{\partial b_R}{\partial z}\hat{\mathbf{R}} + B_0\frac{\partial b_\theta}{\partial z}\hat{\boldsymbol{\theta}} + B_0\frac{\partial b_z}{\partial z}\hat{\mathbf{z}}.\end{aligned} \tag{27.8}$$

So the linearized momentum equation reads

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{1}{2}\Omega u_R \hat{\boldsymbol{\theta}} - 2\Omega u_\theta \hat{\mathbf{R}} = -\frac{1}{\rho}\nabla P + B_0 \frac{\partial b_R}{\partial z}\hat{\mathbf{R}} + B_0 \frac{\partial b_\theta}{\partial z}\hat{\boldsymbol{\theta}} + B_0 \frac{\partial b_z}{\partial z}\hat{\mathbf{z}}. \tag{27.9}$$

The induction equation can be linearized in a similar way:

$$\begin{aligned}
&\nabla \times (\mathbf{v} \times \mathbf{B}) \\
&\quad \to \nabla \times \big[(R\Omega\hat{\boldsymbol{\theta}}) \times (b_R\hat{\mathbf{R}} + b_\theta\hat{\boldsymbol{\theta}} + b_z\hat{\mathbf{z}}) \\
&\qquad\quad + (u_R\hat{\mathbf{R}} + u_\theta\hat{\boldsymbol{\theta}} + u_z\hat{\mathbf{z}}) \times (B_0\hat{\mathbf{z}})\big] \\
&\quad = \nabla \times [-R\Omega b_R\hat{\mathbf{z}} + R\Omega b_z\hat{\mathbf{R}} \\
&\qquad\quad - u_R B_0\hat{\boldsymbol{\theta}} + u_\theta B_0\hat{\mathbf{R}}] \\
&\quad = \nabla \times [(R\Omega b_z + u_\theta B_0)\hat{\mathbf{R}} \\
&\qquad\quad - u_R B_0\hat{\boldsymbol{\theta}} - R\Omega b_R\hat{\mathbf{z}}] \\
&\quad = \left(\frac{\partial u_R}{\partial z}B_0\right)\hat{\mathbf{R}} + \Bigg(R\Omega\frac{\partial b_z}{\partial z} + \frac{\partial u_\theta}{\partial z}B_0 \\
&\qquad\quad - \frac{3}{2}\Omega b_R + \Omega\frac{\partial (R b_R)}{\partial R}\Bigg)\hat{\boldsymbol{\theta}} \\
&\qquad\quad - \frac{1}{R}\left[\frac{\partial (R u_R B_0)}{\partial R}\right]\hat{\mathbf{z}} \\
&\quad = \left(\frac{\partial u_R}{\partial z}B_0\right)\hat{\mathbf{R}} + \left(\frac{\partial u_\theta}{\partial z}B_0 - \frac{3}{2}\Omega b_R\right)\hat{\boldsymbol{\theta}} \\
&\qquad\quad - \frac{1}{R}\left[\frac{\partial (R u_R B_0)}{\partial R}\right]\hat{\mathbf{z}}.
\end{aligned} \tag{27.10}$$

In the last equation we have also used the condition that $\nabla \cdot \mathbf{b} = 0$.

We will consider a small patch of the disk, so that we can use the Fourier (WKB) ansatz:

$$\mathbf{u} = \mathbf{u}_0 e^{\lambda t + \mathrm{i}(k_R R + k_z z)}, \qquad \mathbf{b} = \mathbf{b}_0 e^{\lambda t + \mathrm{i}(k_R R + k_z z)} \tag{27.11}$$

with the additional assumption that $k_z \gg k_R$. Substituting this into the linearized equations of motions yields

$$\begin{aligned}
\lambda u_r - 2\Omega u_\theta &= -\mathrm{i}k_R\frac{P}{\rho} + \frac{\mathrm{i}k_z B_0 b_r}{4\pi\rho}, \\
\lambda u_\theta + \frac{\Omega}{2}u_r &= \frac{\mathrm{i}k_z B_0 b_\theta}{4\pi\rho}, \\
\lambda u_z &= -\mathrm{i}k_z\frac{P}{\rho} + \frac{\mathrm{i}k_z B_0 b_z}{4\pi\rho}.
\end{aligned} \tag{27.12}$$

We can eliminate P/ρ from the last equation and insert it into the first to obtain (in the limit $k_z \gg k_R$):

$$\lambda u_r - 2\Omega u_\theta = \mathrm{i}k_z\frac{B_0 b_r}{4\pi\rho}. \tag{27.13}$$

The linearized induction equation yields

$$\begin{aligned}
\lambda b_R &= \mathrm{i}k_z B_0 u_R \\
\lambda b_\theta &= -\frac{3}{2}\Omega b_R + \mathrm{i}k_z B_0 u_\theta.
\end{aligned} \tag{27.14}$$

Eliminating b_R in the second equation yields

$$\lambda b_\theta = -\frac{3}{2}\frac{\mathrm{i}k_z B_0 \Omega}{\lambda}u_R + \mathrm{i}k_z B_0 u_\theta. \tag{27.15}$$

Eliminating b_θ and b_R in the momentum equations above and introducing $\omega_A^2 = k_z^2 B_0^2/4\pi\rho$ yields

$$
\begin{aligned}
(\lambda^2 + \omega_A^2) u_R &= 2\lambda\Omega u_\theta, \\
(\lambda^2 + \omega_A^2) u_\theta &= \left(-\frac{\lambda}{2}\Omega + \frac{3}{2}\frac{\omega_A^2 \Omega}{\lambda} \right) u_R.
\end{aligned} \tag{27.16}
$$

Introducing

$$
\mu^2 = \lambda^2 + \omega_A^2, \tag{27.17}
$$

the dispersion relation becomes

$$
f(\mu^2) = \mu^4 + \Omega^2 \mu^2 - 4\Omega^2 \omega_A^2 = 0. \tag{27.18}
$$

The instability is present when the second root of $f(\mu^2)$ occurs for $\mu^2 > \omega_A^2$. This occurs if

$$
f(\omega_A^2) = \omega_A^4 - 3\Omega^2\omega_A^2 < 0 \Rightarrow \boxed{\frac{k_z B_0}{\sqrt{4\pi\rho}} = \omega_A < \sqrt{3}\Omega.} \tag{27.19} \tag{27.20}
$$

We find that if k_z is sufficiently small (sufficiently large wavelength of the perturbation), then the mode is unstable. However, if the background field is too large, then the condition in Equation (27.19) might only be satisfied for k_z that corresponds to modes larger than the disk size, so the system is stable.

The growth rate of the instability is found by solving the dispersion relation to find

$$
\lambda^2 = \frac{\Omega}{2}\left(-\Omega + \sqrt{\Omega^2 + 16\omega_A^2} \right) - \omega_A^2, \tag{27.21}
$$

which is maximized when

$$
\omega_A = \frac{\sqrt{15}}{4}\Omega \approx \Omega, \tag{27.22}
$$

which corresponds to

$$
k_{\max} \approx \frac{\sqrt{4\pi\rho}}{B_0}\Omega. \tag{27.23}
$$

The growth rate of the most unstable mode is

$$
\boxed{\lambda_{\max} = \frac{3}{4}\Omega.} \tag{27.24}
$$

It is instructive to check the limit of our analysis when $B_0 \to 0$, which corresponds to $\omega_A \to 0$. If $\omega_A \equiv 0$, then Equation (27.16) reduces to

$$
\lambda^2 u_R = -\Omega^2 u_R, \qquad \lambda^2 u_\theta = -\Omega^2 u_\theta. \tag{27.26}
$$

That is the perturbations oscillate with the epicyclic frequency Ω.

In the limit of $0 < \omega_A \ll \Omega$ (long wavelength perturbations, or weak magnetic field), we can drop all the ω_A terms in Equation (27.16), with the exception of the term multiplied by Ω to obtain

$$
\begin{aligned}
\lambda^2 u_R &= 2\lambda\Omega u_\theta, \\
\lambda^2 u_\theta &= \left(-\frac{\lambda}{2}\Omega + \frac{3}{2}\frac{\omega_A^2 \Omega}{\lambda} \right) u_R.
\end{aligned} \tag{27.27}
$$

Eliminating u_θ, we obtain:

$$
\lambda^2 u_R = \left(\frac{3\omega_A^2\Omega^2}{\lambda^2} - \Omega^2 \right) u_R, \tag{27.28}
$$

which has unstable solutions with $\lambda^2 > 0$ as long as $\omega_A > 0$.

27.3 In-class worksheet

There is a simple mechanical equivalent of the magnetorotational instability (MRI) in a disk with vertical magnetic field. This model is concerned with two particles of mass m orbiting a central object with angular velocity $\Omega = \Omega(R)$ (not necessarily Keplerian). The particles are connected by a spring with spring constant mK, as shown in Figure 27.2.

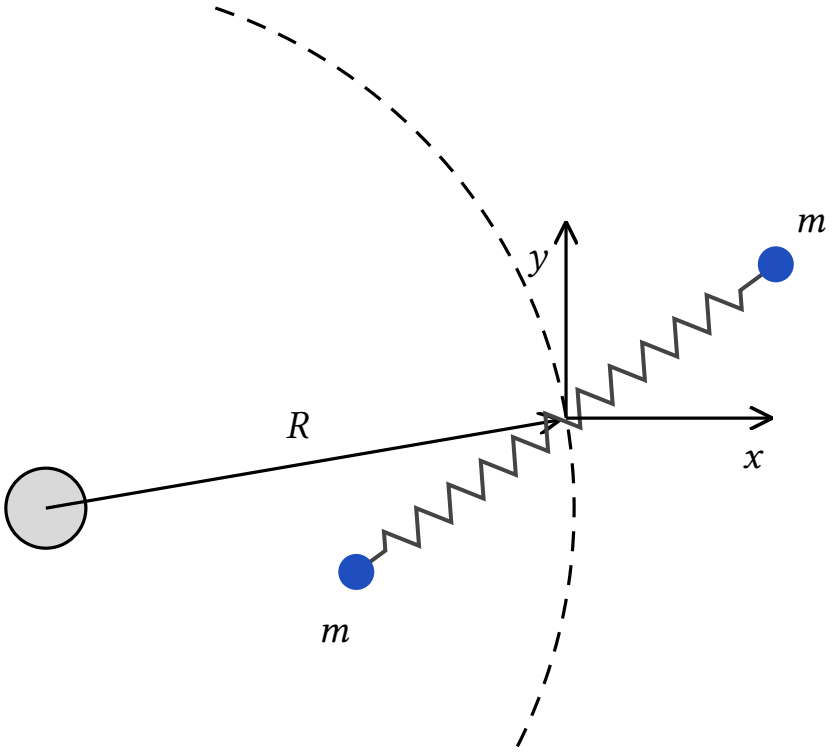


Figure 27.2: Mechanical analogue of the magnetorotational instability: two masses connected by a spring, orbiting a central object.

The coordinates x and y denote the displacement of one of the two masses from the common center of mass.

1. Assuming $0 \leq x, y \ll R$, show that the displacement satisfies the differential equations

$$\begin{aligned} \ddot{x} - 2\Omega\dot{y} &= -\left(K + \frac{\mathrm{d}\Omega^2}{\mathrm{d}\log R}\right)x, \\ \ddot{y} + 2\Omega\dot{x} &= -Ky. \end{aligned} \tag{27.29}$$

   Note that these are the same equations as those obtained for the MRI in a disk threaded by a vertical magnetic field, $k_R = 0$, and $K = (v_A k_z)^2 = \omega_A^2$.

2. Look for solutions in the form $x(t) = x_0 e^{\lambda t}$, $y(t) = y_0 e^{\lambda t}$. Find λ.

3. Under what conditions is the system unstable?

27.4 Further reading

- C. Clarke and B. Carswell [1], Chapter 13, for the MHD equations used here.
- S. A. Balbus and J. F. Hawley [5], the paper that established the MRI as the origin of turbulence in accretion disks.

- S. A. Balbus and J. F. Hawley [6], a comprehensive review of the MRI and angular-momentum transport in disks.

Appendices

A. Vector Calculus

> **Note**
>
> This chapter is in preparation.

Bibliography

[1] C. Clarke and B. Carswell, *Principles of Astrophysical Fluid Dynamics*. Cambridge: Cambridge University Press, 2007.

[2] R. M. Kulsrud, *Plasma Physics for Astrophysics*. Princeton: Princeton University Press, 2005.

[3] E. P. Velikhov, "Stability of an Ideally Conducting Liquid Flowing Between Cylinders Rotating in a Magnetic Field," *Soviet Physics JETP*, vol. 9, p. 995, 1959.

[4] S. Chandrasekhar, *Hydrodynamic and Hydromagnetic Stability*. Oxford: Clarendon Press, 1961.

[5] S. A. Balbus and J. F. Hawley, "A Powerful Local Shear Instability in Weakly Magnetized Disks. I. Linear Analysis," *The Astrophysical Journal*, vol. 376, pp. 214–233, 1991, doi: 10.1086/170270.

[6] S. A. Balbus and J. F. Hawley, "Instability, Turbulence, and Enhanced Transport in Accretion Disks," *Reviews of Modern Physics*, vol. 70, pp. 1–53, 1998, doi: 10.1103/RevModPhys.70.1.